

Auteur: Thomas Eertmans
 Studierichting: Informatica
 Oefeningenreeks 3G nr. 18.2

$$f(x) = \sqrt{\frac{x}{x+1}}$$

$$\text{dom } f : \frac{x}{x+1} \geq 0$$

$$\Rightarrow]-\infty, -1[\cup [0, +\infty[$$

Verticale asymptoten:

$$\lim_{x \rightarrow -1+} \sqrt{\frac{x}{x+1}} \text{ bestaat niet}$$

$$\lim_{x \rightarrow -1-} \sqrt{\frac{x}{x+1}} = +\infty$$

$$\Rightarrow \text{VA: } x = -1$$

Horizontale asymptoten:

$$\lim_{x \rightarrow -\infty} \sqrt{\frac{x}{x+1}} = \sqrt{\lim_{x \rightarrow -\infty} \frac{x}{x+1}} = \sqrt{1} = 1$$

$$\lim_{x \rightarrow +\infty} \sqrt{\frac{x}{x+1}} = 1$$

$$\Rightarrow \text{HA: } y = 1$$

Er zijn geen schuine asymptoten want er zijn al twee horizontale asymptoten.

$$\begin{aligned} f'(x) &= \frac{1}{2\sqrt{\frac{x}{x+1}}} \cdot \left(\frac{x}{x+1}\right)' \\ &= \frac{1}{2\sqrt{\frac{x}{x+1}}} \cdot \frac{1 \cdot (x+1) - x \cdot 1}{(x+1)^2} \\ &= \frac{1}{2\sqrt{\frac{x}{x+1}}(x+1)^2} = \frac{1}{2\sqrt{x}(x+1)^{\frac{3}{2}}} \end{aligned}$$

De eerste afgeleide heeft geen kritieke punten.

$$\begin{aligned} f''(x) &= \frac{1}{2} \cdot \left(-\frac{1}{\left(\sqrt{x}(x+1)^{\frac{3}{2}}\right)^2} \cdot \left(\sqrt{x}(x+1)^{\frac{3}{2}}\right)' \right) \\ &= \frac{1}{2} \cdot \left(-\frac{1}{x(x+1)^3} \cdot \left(\frac{(x+1)^{\frac{3}{2}}}{2\sqrt{x}} + \frac{3\sqrt{x}(x+1)^{\frac{1}{2}}}{2} \right) \right) \end{aligned}$$

$$\begin{aligned}
& \frac{1}{2} \cdot \left(\left(-\frac{1}{x(x+1)^3} \cdot \frac{(x+1)^{\frac{3}{2}}}{2\sqrt{x}} \right) + \left(-\frac{1}{x(x+1)^3} \cdot \frac{3\sqrt{x}(x+1)^{\frac{1}{2}}}{2} \right) \right) \\
& \frac{1}{2} \cdot \left(\left(-\frac{1}{2x^{\frac{3}{2}}(x+1)^{\frac{3}{2}}} \right) + \left(-\frac{3}{2x^{\frac{1}{2}}(x+1)^{\frac{5}{2}}} \right) \right) \\
& = \frac{1}{2} \cdot \left(\frac{-(x+1) - 3x}{2x^{\frac{3}{2}}(x+1)^{\frac{5}{2}}} \right) \\
& = \frac{1}{2} \cdot \left(\frac{-1 - 4x}{2x^{\frac{3}{2}}(x+1)^{\frac{5}{2}}} \right) \\
& = -\frac{1+4x}{4x^{\frac{3}{2}}(x+1)^{\frac{5}{2}}}
\end{aligned}$$

De tweede afgeleide heeft een VA in $x = 0$.

x	-1			0	
$f(x)$	+		///	0	+
$f'(x)$	+	+	///	+	+
$f''(x)$	+	+	///		-
$f(x)$	$\nearrow$	$\nearrow$	///	$\nearrow$	$\nearrow$
	$\smile$	$\smile$	///	$B(0)$	$\frown$

3G-18.2-thomas.eertmans.INF.png

References